

Numerical Analysis

Lab 3

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Method 3.1 (Fixed point method)

A **fixed point method** consists in the sequence

$$x^{(k+1)} = \phi(x^{(k)})$$

with **consistency** and **convergence** properties, that means ϕ is s.t.

$$\xi = \phi(\xi) \quad \text{and} \quad \lim_{k \rightarrow \infty} |x^{(k)} - \xi| = 0.$$

Theorem 3.1 (Local convergence, Ostrowski theorem)

Let ξ be a fixed point of a function ϕ which is continuous and differentiable in a neighborhood of ξ . If $|\phi'(\xi)| < 1$, then there exists $\delta > 0$ s.t., for any $x^{(0)} \in (\xi - \delta, \xi + \delta)$, the sequence $x^{(k+1)} = \phi(x^{(k)})$ converges to ξ . Moreover, the following limit holds

$$\lim_{k \rightarrow \infty} \frac{x^{(k+1)} - \xi}{x^{(k)} - \xi} = \phi'(\xi)$$

Remark 1

Let ξ be a fixed point of the a function ϕ which is continuous and differentiable in a neighborhood of ξ .

- if $|\phi'(\xi)| > 1$, then the sequence $x^{(k+1)} = \phi(x^{(k)})$ will not converge to ξ ;
- if $|\phi'(\xi)| = 1$, then no general conclusion can be drawn both convergence and divergence become possible.

Remark 2 (Geometrical interpretation)

Solving the fixed point problem $x = \phi(x)$ is equivalent to solve the system

$$\begin{cases} y = x \\ y = \phi(x) \end{cases}$$

i.e. to determine the intersections between $\phi(x)$ and the bisector line $y = x$ of the first and the third quadrants.

Theorem 3.2

If ϕ is $\mathcal{C}^p$ in a suitable neighborhood of ξ , and if

$$\phi^{(i)}(\xi) = 0 \quad i = 1, \dots, p-1, \quad \phi^{(p)}(\xi) \neq 0,$$

then the fixed-point method $x^{(k+1)} = \phi(x^{(k)})$ has order p , i.e.

$$\lim_{k \rightarrow \infty} \frac{x^{(k+1)} - \xi}{(x^{(k)} - \xi)^p} = \frac{\phi^{(p)}(\xi)}{p!}$$

Remark 3 (How to estimate the order of convergence p)

Consider the limit of the above-mentioned quantity for two consecutive steps and, for $k \rightarrow \infty$, analyze the corresponding error defining $e^k = x^k - \xi$:

$$\frac{e^{(k+1)}}{(e^{(k)})^p} = \frac{e^{(k)}}{(e^{(k-1)})^p},$$

so that

$$\frac{e^{(k+1)}}{e^{(k)}} = \left(\frac{e^{(k)}}{e^{(k-1)}} \right)^p,$$

and applying the logarithm

$$p = \frac{\log \frac{e^{(k+1)}}{e^{(k)}}}{\log \frac{e^{(k)}}{e^{(k-1)}}} = \frac{\log e^{(k+1)} - \log e^{(k)}}{\log e^{(k)} - \log e^{(k-1)}}.$$

Exercise 3.1

Solve the equation $x^2 - 5 = 0$ to find the root $\xi = \sqrt{5}$ using the following iterative methods:

- ① $x^{(k+1)} = 5 + x^{(k)} - (x^{(k)})^2;$
 - ② $x^{(k+1)} = \frac{5}{x^{(k)}};$
 - ③ $x^{(k+1)} = 1 + x^{(k)} - \frac{1}{5}(x^{(k)})^2;$
 - ④ $x^{(k+1)} = \frac{1}{2} \left(x^{(k)} + \frac{5}{x^{(k)}} \right).$
- a Are these iterations consistent and convergent? Motivate your answer.
 - b Implement a function for the generic fixed point iteration with function ϕ .
 - c Assess numerically the convergence of the proposed iterative methods.

Exercise 3.2

Let

$$f(x) = x^3 - \sin(\pi x); \quad a = -1/2, \quad b = 1/2, \quad x^{(0)} = 0.5$$

and apply the following fixed-point method

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{\frac{f(b) - f(a)}{b - a}}$$

(known as the chord method) to solve $f(\xi) = 0$

- a does the method converge? how many iterations does it take to converge?
- b consider $x^{(0)} \neq 0.5$, will the method converge $\forall x^{(0)} \in [a, b]$?

Exercise 3.3

Let

$$f(x) = x^3 + \sin(\pi x); \quad a = -1/2, \quad b = 1/2, \quad x^{(0)} = 0.5$$

and apply the following fixed-point method

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{\frac{f(b) - f(a)}{b - a}}$$

(known as the chord method) to solve $f(\xi) = 0$

- a does the method converge? how many iterations does it take to converge?
- b consider $x^{(0)} \neq 0.5$, will the method converge $\forall x^{(0)} \in [a, b]$?

Exercise 3.4

Consider the following fixed point iteration

$$\begin{cases} x^{(k+1)} = g(x^{(k)}) \\ g(x) := \frac{1}{2} \left(x + \frac{a}{x} \right) \end{cases}$$

which one of the following statements is **true**

- ☐ a the iteration converges $\forall x^{(0)} \in \left(0, \frac{1}{a}\right)$
- ☐ b $x = \frac{1}{a}$ is a fixed point for g
- ☐ c $x = \sqrt{a}$ is a fixed point for g

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2 (Modified) Newton Method

Method 3.2 (Newton method)

Newton method consists in approximating the solution of $f(x) = 0$ with the sequence

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})} \quad \text{with} \quad k \geq 0 \quad \text{and} \quad f'(x^{(k)}) \neq 0$$

Modified Newton method II

If the root ξ is not simple the Newton method converges with order one. If m is the multiplicity of the root, then the modification

$$x^{(k+1)} = x^{(k)} - m \frac{f(x^{(k)})}{f'(x^{(k)})} \quad \text{with} \quad k \geq 0 \quad \text{and} \quad f'(x^{(k)}) \neq 0$$

allow us to recover the second order of convergence.

Newton method as fixed point method I

The Newton method can be written as a fixed point method

$$\phi(x) = x - \frac{f(x)}{f'(x)}$$

if $f'(\xi) \neq 0$ (ξ is a simple root) then

$$\phi'(\xi) = 0 \quad \text{and} \quad \phi''(\xi) = \frac{f''(\xi)}{f'(\xi)} \neq 0$$

so that, exploiting the previous theorem, it can be deduced that Newton method has order 2.

Newton method as fixed point method II

If $f'(\xi) = 0$ (ξ is a multiple root) of multiplicity m (i.e. $f^{(i)}(\xi) = 0$ for $i = 1, \dots, m-1$ and $f^{(m)}(\xi) \neq 0$), then without modification the Newton method has order 1.

$$\phi'(\xi) = 1 - \frac{1}{m} \neq 0$$

if m is known, the Newton method can be modified

$$\phi(x) = x - m \frac{f(x)}{f'(x)}$$

so that the modified Newton method has order 2 again.

Exercise 3.5

Let

$$f(x) = 1 + x^2 - \cos(s)$$

and apply the Newton-Raphson method to solve $f(\xi) = 0$

- a will the method converge for $x^{(0)} \in (-1, 1)$?
- b estimate *a posteriori* the order of convergence of the method
- c modify the method in order to increase the order of convergence

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3 Homework

Homework 3.1 (Fixed point method)

- a Let $\phi(x) = x - x^3$, which admits $\xi = 0$ as a fixed point. Compute $\phi'(\xi)$ and test the convergence of the sequence $x^{(k+1)} = \phi(x^{(k)})$ for $x^{(0)} \in [-1, 1]$.
- b Let $\phi(x) = x + x^3$, which admits $\xi = 0$ as a fixed point. Compute $\phi'(\xi)$ and test the convergence of the sequence $x^{(k+1)} = \phi(x^{(k)})$ for $x^{(0)} \in [-1, 1]$.

Homework 3.2 (Fixed point method, Newton method)

Given the equation

$$f(x) = e^{x^2} \log(x + 1) = 1$$

consider the following three iterative methods

- ❶ $x^{(k+1)} = \sqrt{-\log \log(x^{(k)} + 1)}$;
- ❷ $x^{(k+1)} = x^{(k)} e^{(x^{(k)})^2} \log(x^{(k)} + 1)$;
- ❸ Newton method.

Verify the consistency of the schemes and apply them to compute the root ξ with a tolerance equal to 10^{-3} . Comment on the results.

References I